

Project Proposal

Smart Room Monitoring Sensor Node with Event-Triggered Interrupt Support

1. Abstract

This project proposes the design and implementation of a sensor node platform for indoor room monitoring. The system combines two digital sensors and one analog sensor to monitor environmental conditions and detect physical interaction events. The selected sensors are a BME680 environmental sensor for temperature, humidity, pressure, and gas-related air quality data, a VEML7700 ambient light sensor for light intensity measurement, and an FSR (force-sensitive resistor) as an analog sensor for touch/pressure event detection. The sensor node communicates with a Zephyr-based base station through a custom communication interface. On the base station side, we will implement a Zephyr driver that exposes the sensor node through a structured interface, together with a demonstration application. To target the highest grade, the project will include runtime configuration support, an interrupt line between the sensor node and the base station, and a structured testing strategy.

2. Motivation

The goal of the project is to build a complete embedded sensing platform rather than only reading isolated sensors. The proposed system is relevant because room monitoring is a realistic use case for embedded systems and allows us to combine analog sensing, digital sensing, communication protocol design, Zephyr driver development, and event-driven system behavior in one coherent project. This project also fits the course goal of practicing low-level programming, debugging, testing, and system integration.

3. System Overview

The system consists of two parts:

Sensor node

The sensor node is responsible for:

- reading the BME680 sensor

- reading the VEML7700 sensor
- reading the analog value from the FSR through ADC
- packaging sensor data into protocol frames
- detecting threshold-based pressure events
- generating an interrupt signal to the base station when an event occurs

Base station

The base station is responsible for:

- communicating with the sensor node
- requesting sensor values
- receiving event notifications
- exposing the sensor node through a Zephyr driver
- running a demo application for visualization and testing

4. Hardware Design

Selected sensors

- **BME680**: digital environmental sensor measuring temperature, humidity, pressure, and gas/VOC-related air quality values.
- **VEML7700**: digital ambient light sensor with I²C interface.
- **FSR**: analog force-sensitive resistor used for pressure/touch detection. The FSR changes resistance depending on applied force and will be connected in a voltage-divider circuit to an ADC input.

Signal types

- BME680: digital
- VEML7700: digital
- FSR: analog

This satisfies the course requirement of including at least one analog sensor.

Circuit outline

- BME680 connected via I²C

- VEML7700 connected via I²C
- FSR connected to an ADC pin using a resistor divider
- one GPIO line reserved as an interrupt line from sensor node to base station
- pull-up resistors for I²C if required
- decoupling capacitors for stable operation

5. Software Design

Sensor node firmware

We plan to implement the sensor node firmware using the Pico SDK. Its responsibilities are:

- periodic sampling of all sensors
- ADC acquisition for the FSR
- local threshold comparison for event detection
- frame-based communication with the base station
- interrupt generation when the FSR exceeds a configurable threshold

Communication interface

We propose to use **UART** as the communication interface between sensor node and base station.

Reasons:

- simple and robust for debugging
- easy to log and inspect during development
- suitable for a custom packet-based protocol

Protocol concept

The protocol will support commands such as:

- READ_ALL
- READ_ENVIRONMENT
- READ_LIGHT
- READ_FORCE

- SET_SAMPLING_RATE
- SET_FORCE_THRESHOLD
- ENABLE_INTERRUPT
- CLEAR_EVENT

Responses will contain:

- frame header
- command ID
- payload length
- payload
- checksum/CRC

Base station software

The base station will use Zephyr. We will implement a custom driver that:

- retrieves data from the sensor node
- provides access to multiple measured quantities
- supports configuration of sampling rate and event threshold
- handles interrupt-based event notification

We will also provide a demonstration application showing how to read values and react to events.

6. 5-Grade Features

To explicitly target grade 5, the project includes:

Three-sensor platform

The system uses three sensors, exceeding the minimum for grade 4.

Driver configuration

The Zephyr driver will support runtime configuration, for example:

- sampling interval
- event threshold
- interrupt enable/disable

This addresses the "configuration in driver interface" requirement for grade 4 and above.

Interrupt line

The sensor node will assert a dedicated interrupt GPIO when a pressure event is detected. The base station will then fetch detailed data over UART. This directly targets the grade 5 requirement of including an interrupt line between the sensor node and the base station.

Testing strategy

We will implement a structured testing strategy, including unit, integration, protocol, and end-to-end tests. This directly targets the grade 5 requirement of a good testing strategy and implementation.

7. Technical Obstacles and Planned Solutions

Obstacle 1: Analog sensor nonlinearity and noise

The FSR output is nonlinear and sensitive to noise.

Planned solution:

We will calibrate the signal experimentally, apply filtering/averaging, and use threshold hysteresis to avoid unstable event triggering.

Obstacle 2: I²C multi-device integration

Using two digital sensors on the same bus may introduce address or wiring problems.

Planned solution:

We will verify sensor addresses early, test one device at a time first, and add pull-up resistors if needed.

Obstacle 3: Communication robustness

UART data may be corrupted or incomplete.

Planned solution:

We will implement a frame format with checksum/CRC, timeout handling, and basic resynchronization logic.

Obstacle 4: Driver abstraction

The custom sensor node must be cleanly represented on the Zephyr side.

Planned solution:

We will define a small, stable protocol and map measurements into a clear driver interface, testing each data path separately.

Obstacle 5: Event synchronization

An interrupt may occur while the base station is already reading values.

Planned solution:

The sensor node will latch the event state until the base station acknowledges it.

8. Time Plan

Week 1

- finalize sensor selection
- complete wiring diagram
- set up development environments
- verify basic communication with each sensor

Week 2

- implement BME680 and VEML7700 reading
- implement FSR ADC reading
- test analog voltage range and threshold behavior

Week 3

- design and implement UART protocol
- test frame encoding/decoding
- connect node and base station

Week 4

- implement Zephyr driver

- implement basic data retrieval commands
- develop demo application

Week 5

- add configuration support
- add interrupt line and event handling
- perform integration testing

Week 6

- perform structured testing
- finalize code
- prepare final presentation and report